Analyzing STAAR Data

Dancing the TEA Two-Step

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"He who has a why to live for can bear almost any how."
— *Friedrich Nietzsche*
(Twilight of the Idols)

My aim here is to contribute something to the improvement of learning in our school district; or in any school district. My basic supposition is that Boards of Trustees will welcome information designed to help them create rational policy action aimed at improving **future** learning in their district.

BACKGROUND

The Texas Education Agency (TEA) collects data from every school in Texas, performs various calculations, and produces reams of reports and tables summarizing their results by academic year. These reports consist primarily of calculations mandated by the state legislature, including those designed to allow districts to fulfill their public notification requirements.

These statistical summaries are available to the public and the school districts. Unfortunately for the public and the school districts, the information is not nearly as helpful as it could be. In fact, it takes so much 'data wrangling' to put it into a form that is remotely helpful for assessing and improving district and campus learning that it is simply not done.

Instead, administrators and school boards stare at endless rows and columns of mind-numbing data fragments that are 'sliced and diced' to an

extent that even RonPopeil the inventor of the Chop-O-Matic would be envious!

Here's a partial example from a TEA report called 'Snapshot':

STAAR by Subject

Percent of Tests	All Subjects	ELA Reading	Mathematics	Science	Social Studies
At Approaches Grade Level Standard or Above	80%	79%	78%	84%	85%
At Meets Grade Level Standard or Above	53%	55%	48%	58%	60%
At Masters Grade Level Standard	25%	23%	21%	29%	39%

The entire report can be found here:
PDF

(HTML)

And, here's a partial example from a TEA report called 'School Report Card:

STAAR Outcomes

This section provides STAAR performance and Academic Growth outcomes. Academic Growth measures whether students are maintaining performance or improving from year to year

		State	District	Campus	African American	Hispanic	White	American Indian	Asian	Pacific Islander	Two or More Races	Econ Disadv
STAAR Performance Rates at Approaches Grade Level or Above (All Grades Tested)												
All Subjects	2022	74%	80%	**87%**	84%	86%	90%	89%	100%	71%	89%	80%
	2021	67%	78%	**89%**	85%	87%	92%	100%	100%	93%	96%	83%
ELA/Reading	2022	75%	79%	**81%**	74%	80%	84%	*	100%	50%	84%	72%
	2021	68%	77%	**85%**	80%	82%	88%	100%	100%	83%	92%	78%
Mathematics	2022	72%	78%	**86%**	86%	84%	88%	*	100%	*	83%	81%
	2021	66%	79%	**89%**	86%	85%	94%	*	100%	*	96%	84%
Science	2022	76%	84%	**95%**	94%	94%	97%	*	100%	*	98%	91%
	2021	71%	83%	**94%**	90%	93%	95%	-	100%	*	98%	89%
Social Studies	2022	75%	85%	**95%**	91%	94%	97%	*	100%	*	100%	89%
	2021	73%	87%	**95%**	93%	94%	97%	*	100%	*	100%	90%

The entire report can be found here:
PDF

(HTML)

THE PROBLEM

So, what is a school district to make of these mountains of numbers? Do they help districts improve learning in the future? In their present form, the answer is a resounding "NO!"

The 'Snapshot' offered by the TEA is a good example. My hat is off to their programmers. It is an impressive compilation of statistics all in one place. It is well-suited for the needs of the TEA and their reporting responsibilities to the legislature; but completely ill-suited for helping the concerned school board, superintendent, or tax-paying citizen make an analysis aimed at improving future learning.

Why, then, are they 'ill-suited' for district improvement efforts?

First, many of the TEA categories are highly aggregated combinations of similar categories. This permits TEA to render abstract judgements about a district or campus. One such judgement is called an "Accountability Rating". It is well and truly from the '30,000 foot level'. This lack of specificity renders these reports unsuitable for understanding any specific categories or levels your board may be interested in.

By analogy, it's like trying to repair your car by reading the overall repair records of similar models. What you need is diagnostic data about what is happening (or not happening) with your specific vehicle.

Here is an example of aggregated STAAR categories in a TEA report:

STAAR by Subject

Percent of Tests	All Subjects	ELA Reading	Writing	Mathematics	Science	Social Studies
At Approaches Grade Level Standard or Above	78%	77%	63%	79%	83%	87%
At Meets Grade Level Standard or Above	52%	52%	30%	50%	60%	65%
At Masters Grade Level Standard	23%	20%	6%	23%	31%	43%

Second, the TEA reports are siloed by academic year making it nearly impossible to create a time-period-by-time-period data frame of any categories or levels the board may wish to analyze. In fact, without time-period-by-time-period data ***no analysis is possible.***

Through an embarrasing amount of time spent 'data wrangling' I was able to construct from the 'Snapshot' resports a small time-period-by-time-period data frame for STAAR scores for the subject-matter categories at the district-level for the academic years ended 2016-2022. (No STAAR scores were available for the 2019 academic year.)

Here we see a comprehensive view of all subject categories for one level of achievement for all years in the sample:

STAAR by Subject

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All data I've seen so far as a board member are descriptive only, with no analysis present.

THE SOLUTION

For improvement, we must move from a 'snapshot' point of view to an 'analysis' point of view. That means the data we seek must include the context of time. All improvement work requires the context of time.

The important information for understanding district capability, and the prediction of future performance, is contained in the time-period-by-time-period data frame. The order of the data matters. The more periods included in the analysis, the stronger the evidence for your inferences.

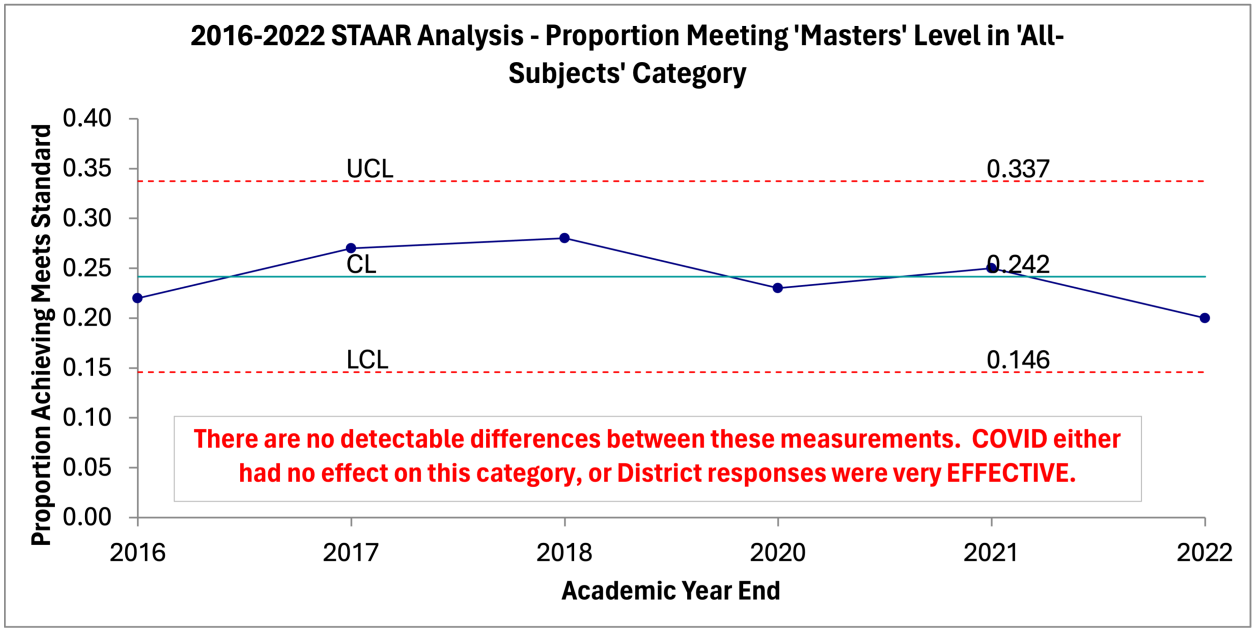
Humans naturally seek context. That is where notions such as 'this month vs last month vs YTD vs a year ago this month' arose. Those are insufficient for analysis, but they soothe our need for context of some sort.

Analysis, on the other hand, allows us to answer the important questions:

- These numbers vary from period to period. Does that tell us anything important?
- Are the differences due to random fluctuations or some strong influence now present in our system?

- Do these different numbers constitute a trend (good, or bad)?
- The legislature (via the TEA) declares that certain average STAAR scores 'meet requirements.' What is the **demonstrated** capability of our district to deliver those STAAR scores? In other words, what is rational to expect from our system in the future?
- Should we continue this approach, abandon it, or continue it with some tweaks? Like all bureaucracies, school districts often find that practices, policies and procedures are seldom abandoned. They just become 'the way we do it here.'

Here is an example of what analysis of some STAAR data looks like:



The TEA is charged with preparing the required reports, and they do. Boards of Trustees are charged with **managing** and **improving** their districts.

Managing a district consists of budgeting, administration, and resource allocation. Essential, to be sure.

Budgeting is based on (hopefully) very careful calculations of estimated revenue and expenses expected for the next academic year. Resource allocation within and between budget categories are determined by the priorities and risk tolerance of the Board of Trustees.

Resources for facility expansion and improvement depend on both the

demographics of the district and the vision of the Executive team.

District improvement is a totally different exercise. To improve, boards must be able to assess the demonstrated capability and effectiveness of learning in their district. .

The context of **TIME** is crucial to assess the capability and effectiveness of any system or process, including learning.

Nearly every district has a 'reputation' (for better or worse) in the region, state, or nation. Some districts have reputations for great athletic programs, some for music/arts, some for academics. Those reputations are built over time.

STAAR scoring by subject matter (from the 2022-2023 Snapshot) To view entire report from pdf format click here: pdf To view entire report from html format click here:
(html)

STAAR by Subject

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At Masters Grade Level Standard	23%	20%	6%	23%	31%	43%

The section reporting STAAR scoring by ethnicity:

STAAR by Ethnicity

Percent of Tests	African American	Hispanic	White	American Indian	Asian	Pacific Islander	Two or More Races	Economically Disadvantaged
At Approaches Grade Level Standard or Above	75%	75%	83%	80%	88%	81%	82%	67%
At Meets Grade Level Standard or Above	46%	46%	60%	46%	68%	51%	56%	37%
At Masters Grade Level Standard	18%	20%	29%	26%	34%	16%	25%	15%

School boards deal with two types data:

- Enumerative: quantitative information limited to a description of the past, usually for compliance purposes.
- Analytic: data subjected to analysis for estimating a causal explanation of phenomena observed in the past, with the aim of improvement in the future.

TEA produces enumerative data estimating the state of affairs at one point in the past. The aim of such data collection is to comply with legislative requirements. The purpose is to take action ON the district; reward or punishment either by the state or the public.

An analytic study focuses on predicting future results of policy actions taken now. The aim of such data analysis is to take action IN the district.

The aim of a TEA report is descriptive. How many or how much. They are estimates of how many people belong to this or that category for a specified academic year. Or, to report the financial aspects of operations during the school year. Their aim is NOT to find out WHY there are so many or so few

in this or that category: merely how many.

School boards should look to take action on the district-level processes, or cause-system, that produced the data described in the TEA reports, the aim being to improve learning in the future. Interest centers on future learning, not the learning of the past. For example : adopt Policy B over A, or hold on to Policy A, or continue the study of Policy B.

Enumerative study: A statistical study in which action will be taken on the material in the frame being studied.

Analytic study: A statistical study in which action will be taken on the process or cause-system that produced the frame being studied. The aim being to improve practice in the future.

In other words, an enumerative study is a statistical study in which the focus is on judgment of results, and an analytic study is one in which the focus is on improvement of the process or system which created the results being evaluated and which will continue creating results in the future. A statistical study can be enumerative or analytic, but it cannot be both.

Statistical theory in enumerative studies is used to describe the precision of estimates and the validity of hypotheses for the population studied. In analytical studies, the standard error of a statistic does not address the most important source of uncertainty, namely, the change in study conditions in the future.

Deming's philosophy is that management should be analytic instead of enumerative. In other words, management should focus on improvement of processes for the future instead of on judgment of current results.

That 27% must in part depend on chance. If we imagine a set of constant conditions, Statistics and Reality (draft) Page 2 Nov 21, 1998 which would lead on average to 100 murders, we can, on the simplest mathematical model, expect the number we actually see to be anything between 70 and 130.

If there were 130 murders one year, and 70 the next, many people would think that there had been a great improvement: but this could be just chance. So the first question we could ask is, "Could the 27% reduction be due to chance?"

That is the least of our problems. The murders may be related, as in a

war between drug barons. If so, the model is wrong, since it assumes that murders are independent. Or the methods of counting might have changed from one year to the next (are you counting all suspicious deaths, or only cases solved?). Without knowing about such things we cannot predict from these figures to what will happen next year. So if we want to draw the conclusion that the 27% reduction is a "real" one, that is, one which will continue in the future, we must use knowledge about the problem that is not given by those figures alone. Even less can we predict accurately what would happen in a different city, or a different country. The causes of crime, or the effect of a change in policing methods, may be completely different.

Then your actions on the basis of the information available become much more effective. Even more, your actions to get more information improve, because when you understand the sources of uncertainty, you understand how to reduce it.

It seems that it might help if we could look at, say, six years of the 'All Subjects' category at one time.

Well, here it is:

When it comes to improving the system of education in a district this view of the data has a BIG PROBLEM…it aggregates more than one rating level into a category. It combines the 'At Approaches' AND the 'At Meets' AND the 'At Masters' rating levels all into one combined total and calls it "At Approaches AND HIGHER Level".

To make that category useful for improvement efforts we have to break it into 3 separate categories so each stands on its own. Just for fun, I call this dilemma 'the TASB Two-Step' - because it will take at least two additional steps to make the category measurements useful. Then, those two tedious steps have to be repeated over each of the five remaining subject-matter categories.

Well, there's nothing for it but to get to work. Here they are,

See Fig. 1

If we try to look at the data over the last 6 years we still have the problem of the 'TASB Two-Step'!

As a board, we should focus first on the combined results of all subjects

tested. This, purportedly, is the performance of the district taken as a whole:

The next rating category also contains more than one rating level. It combines the 'At Meets' AND the 'At Masters' into one combined total.

This rating category is the exception; it contains only one rating level. It shows ONLY the 'At Masters' level totals.

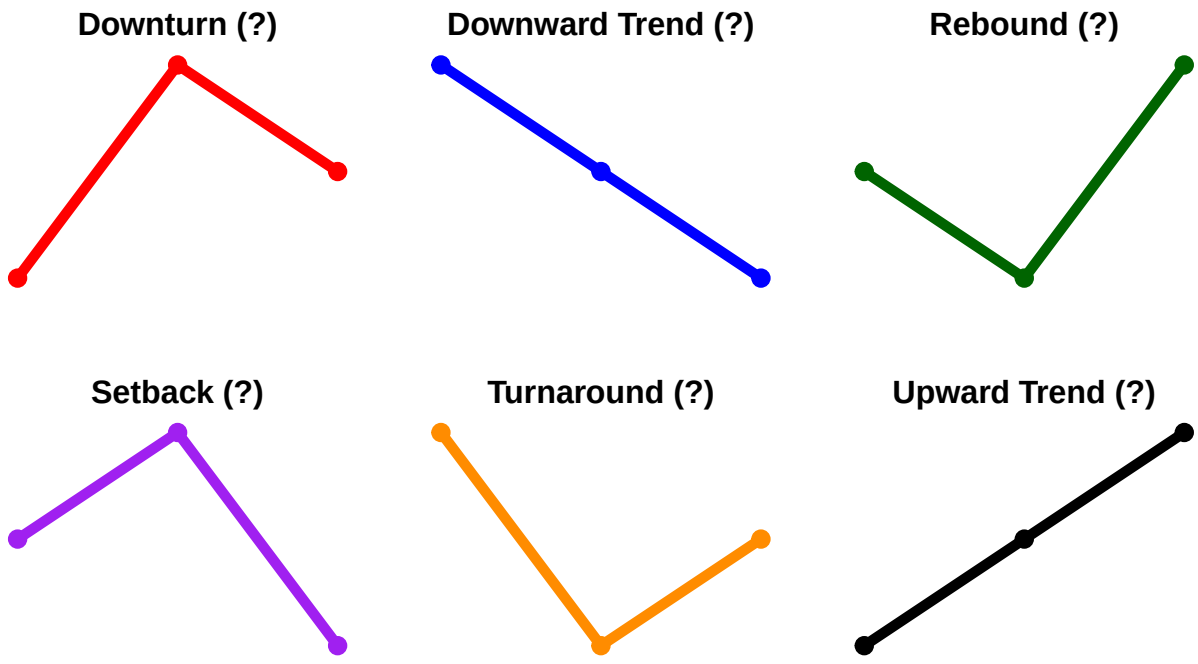

Figure 1: Six Random 'Trends'!